

Sketch-Note Masterclass:

CFA Level II Credit Analysis Models (Module 4)

The Building Blocks of Credit Risk

Expected Loss (EL)

$$EL = POD \times LGD$$

↖ The core metric of credit risk.

Loss Given Default (LGD)

$$LGD = \text{Exposure at Default (EAD)} \times (1 - \text{Recovery Rate})$$

↳ RR varies by industry (e.g., Energy vs. Environmental) and seniority.

Recovery Rate (RR)



"Loss Severity"
(1 - RR)

The Probability Path

Hazard Rate (Conditional POD)

↳ Probability of default in a specific period, given survival until then.

Probability of Survival (POS)

$$POS = (1 - h)^n \text{ (for constant } h)$$

↳ Cumulative POD = 1 - POS

Credit Migration & Transition Matrices

	AAA	AA	A	BBB
AAA				
AA				
A				
BBB				

Transition Matrix

↳ Probabilities of a bond moving from one rating to another.

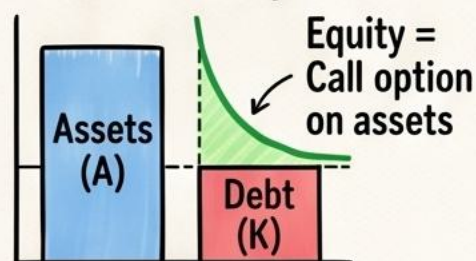
Expected Return Impact

$$\Delta\% \text{ Price} \approx -(\text{Modified Duration} \times \Delta\text{Spread})$$

↳ A to BBB Downgrade: Spread increases; price drop is Duration × spread widening.

Structural vs. Reduced-Form Models

Structural Models (The "Why")



Merton Model.
Default if Asset Value < Debt Barrier ($A < K$).

↳ Weaknesses: Structural (unobservable assets), Reduced-form (lacks economic "why").

Reduced-Form Models (The "When")



Poisson. Uses macro/firm variables to estimate default intensity.

Common Exam Traps



Trap 1: The "Constant" Hazard Rate

↳ $POS_{Yr2} = (1 - POD_1) \times (1 - POD_2)$
↳ Don't assume constant POD.

Trap 2: LGD Units

↳ \$ or % of par? Check units.

Trap 3: Spread Drivers

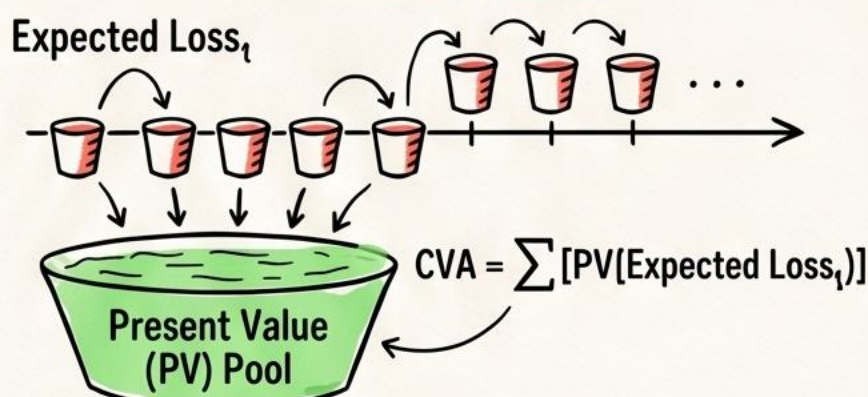
$$\text{Spreads} = \boxed{\text{Credit Risk}} + \boxed{\text{Liquidity Risk}} + \text{Tax}$$

Bond Valuation & CVA

The CVA Identity

$$PV(\text{Risky Bond}) = PV(\text{Default-free Bond}) - CVA$$

Calculating CVA

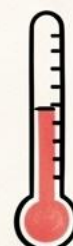


↳ Risk-neutral PODs (for CVA) > actual PODs (include risk premium).

Market Spreads & Liquidity Barometers



TED Spread
3-month USD LIBOR
- 3-month T-Bill
Rising = withdrawing liquidity



Libor-OIS Spread
Bank risk,
OIS = risk-free rate



I-Spread
YTM - Swap rate
for same maturity